VISHWAS · SIH26079 · TEAM HELLFIRE LABS

Devansh

Model and evaluation

Final prep — script, judge Q&A, and every number traced

B.Tech | Slide 2 (second half) · Slide 3 (second half) · Slide 6 (closing card) | about 2 minutes of speaking, spread over three slides

YOUR JOB

You own the result and the build. Every number describing how well VISHWAS works is yours to defend, every question about how the data was fetched and shaped now comes to you, and most technical questions will land on you regardless of which slide is on screen. You are also Mahika's net — see section 3. Your job on stage is to be the person who does not overclaim; your job off it is to make sure she never needs you.

1 RUNNING ORDER

Mahika leads. Your rows are highlighted.

#	SPEAKER	WHAT HAPPENS	TIME
1	Mahika	Slide 1 — opens. Team, problem statement, the hook.	~40s
2	Arnav	Slide 2 — what VISHWAS is, and why the forecast underneath decays.	~70s
3	**Devansh**	**Slide 2 — the measurement, and the result.**	**~60s**
4	Ankit	Slide 3 — opens it: what an ensemble is, and why we score against rain gauges.	~35s
5	**Devansh**	**Slide 3 — the pipeline, the stack, features, label, model, the map.**	**~75s**
6	Mahika	Bridge into slide 4.	~10s
7	Paras	Slide 4 — feasibility, the numbers, the three defects.	~65s
8	Arnav	Slide 4 — viability, the four risks, the precision choice.	~40s
9	Mahika	Slide 5 — the three interviews.	~45s
10	Dhruv	Slide 5 — who it serves, MoES's numbers, the benefits. Then slide 6 — sources.	~75s
11	**Devansh**	**Slide 6 — our own measurement.**	**~20s**
12	Mahika	Closes. The tagline, then into questions.	~20s

2 THE STORY — EVERYONE MUST BE ABLE TO SAY THIS

If a judge stops you in the corridor. Thirty seconds. Learn the shape, not the words.

Every weather forecast India issues arrives with no honesty label.

Forecasters judge confidence today by ensemble spread — run the model 11 times from slightly different starting conditions, and if the runs disagree, distrust the forecast. We measured that method against India's own rain gauges, across all four seasons and three years.

Compared crudely it looks strong: ensemble spread appears to explain 43.9% of the variance in forecast error. Compared fairly — within the same rainfall conditions — it explains under 1%.

So we built the replacement. On that same fair comparison it scores 0.565 and VISHWAS scores 0.728, on a test year the model had never seen.

Most teams pitch a hypothesis. We pitch a measurement.

3 BACKING UP MAHIKA — PRIVATE

Mahika leads. You are the net she should never need. This section is for you and her only — nobody else needs to know it exists, and none of it should be visible to the judges at any point.

THE RULE THAT MATTERS MOST

- × Do not move on a pause. Silence under about five seconds reads as composure to everyone except the person standing in it — she will feel like she is drowning at two seconds while the room hears a considered pause.
- × If you rescue her visibly, the judges see it, and she stops being the lead for the rest of the presentation. A lead who gets saved once is no longer leading. Stepping in early costs far more than three seconds of silence.

THE ONLY TWO TRIGGERS

- × She looks at you. That is the agreed signal and it means take it now.
- × Or the silence passes roughly five seconds with no sign of recovery.
- × Nothing else. Not a stumble, not a lost word, not a wrong number, not a sentence that trails off and restarts. Those are all normal and the room does not register them.

HOW TO TAKE IT, IF YOU HAVE TO

- × Do not acknowledge that anything happened. No “let me take that”, no glance, no apology. Start the next idea as though it was always yours.
- × Use the handoff line she was heading for — every one of them is printed in her pack and in the running order, so you always know where she was going.
- × Give it back at the next slide boundary, by name: “Mahika, the field research.” That restores her without commentary.

NEVER CORRECT HER MID-SENTENCE

- × If she says a figure wrong, let it go. A public correction from a teammate costs far more than the wrong number did, and it tells the judges to start checking everything.
- × Fix it only if a judge raises it, and fix it as information rather than as a correction: “The figure is X — worth being precise about, because it is the one people misread.”

WHERE YOU ARE ACTUALLY WORTH MOST: Q&A

- × That is where pressure genuinely bites, it is where the technical questions land anyway, and backing her up there is invisible to the room.

- ✕ She routes questions by name. If she hesitates on the routing, take it yourself — the judges just see a team answering, not a lead stalling.
- ✕ If a question is aimed at someone who cannot answer it, take it without waiting to be handed it. That reads as a team that knows its own work.

REHEARSE ONE DELIBERATE FREEZE

- ✕ Have her stop dead mid-sentence in practice, on purpose, and hold it. Both of you need to know what five seconds actually feels like and what the recovery sounds like.
- ✕ It is the highest-value ten minutes in your whole preparation — worth more than another read-through of the numbers.

ONE THING ABOUT YOUR OWN DELIVERY

- ✕ Your strongest asset in that room is not confidence. It is that you are the person who found three defects in his own pipeline and put them on a slide.
- ✕ Confidence invites a challenge. “We caught this ourselves” does not.

4 WHAT CHANGED IN YOUR SCRIPT

Your wording is your own. Only these points moved.

- DECK CHANGED — slide 3 now carries the console. The 23-feature figure has come off the slide and in its place is a screenshot of the live dashboard: one gridpoint, its bust probability, and the three reasons it was flagged. You have one new line for it. It matters more than its length suggests: the dashboard and API are Expected Outcome 5 of the problem statement and the explanation panel is Outcome 4, and until now the deck showed neither.
- SAY PROTOTYPE, AND SAY IT RUNS LOCALLY. It is not deployed as a public service: no rate limiting, no caching, and it loads 651,040 rows at startup. Claiming a live service would be false, and it is the kind of claim a judge checks.
- Your section is now about 40 words longer, roughly fifteen seconds. Nothing else in your script changed — the 23 features, the map, the ablation and the slide 6 card are all as they were.
- Your cues updated. Arnav now ends on “Ab Devansh aapko batayenge ki humne us evidence ko kaise measure kiya”, and Ankit now ends on “Gauges are the harder target — and they are the honest one.”
- “Ankit will show you how we built it” reworded. Ankit sets up two concepts; you do the building, so that line was pointing at the wrong thing.
- Dhruv no longer says “Thank you” before your Slide 6 card — his last words are now your clean cue.
- Nothing else. Yours was the cleanest script of the six: every number correct, “variance” said properly, and 651,040 and 1.3 million correctly kept apart.

5 YOUR SCRIPT

Grey boxes are stage directions — do not say them. Lines tagged TRIM are safe to cut if you are running long.

SLIDE 2 — SECOND HALF

► *Arnav ends on “Ab Devansh aapko batayenge ki humne us evidence ko kaise measure kiya.” Enter immediately. No awkward pause.*

> Exactly. And this is what makes VISHWAS different.

> Humne assume nahi kiya that the current method is weak. We measured it — across all four seasons and three years, against India's own gauge observations.

► *PAUSE*

When everything is compared together, ensemble spread appears to explain 43.9% of the variance in forecast error.

That sounds strong.

But much of that relationship comes from geography — places with heavier rainfall naturally tend to have both higher spread and higher errors.

So humne like with like compare kiya — forecasts under the same rainfall conditions.

▸ *PAUSE*

And that 43.9%… collapses to under 1%.

▸ *BIG PAUSE — look at the judges.*

So we're not simply automating an existing judgement. Hum us judgement ko an evidence base de rahe hain.

Then we built the replacement.

On the same fair comparison, ensemble spread scores 0.565. VISHWAS scores 0.728.

▸ *PAUSE*

That's a margin of +0.13 to +0.16, with the 95% interval entirely above zero, on a test year the model had never seen.

Ankit will set up two things you'll need before I show you how it's built.

SLIDE 3 — AFTER ANKIT

▸ *Ankit ends on "Gauges are the harder target — and they are the honest one." Take over straight away.*

And from there, VISHWAS runs through six stages.

Aur importantly, this isn't just a PPT diagram — every stage is a real working script in our repository.

We fetch GEFS reforecast data from NOAA, convert the 11 ensemble members into weather means and spreads, combine that with the gauge truth Ankit just explained, and then label, model and evaluate.

▸ *PAUSE*

Overall, we processed 50.3 GB of data and labelled around 1.3 million gridpoint-days.

The model uses 23 features, all from GEFS and none from the truth data.

Yeh separation important hai — because if truth leaks into the model, the evaluation becomes meaningless.

▸ *PAUSE*

We define a forecast bust when the error between forecast and observed rainfall crosses the 95th percentile, calculated separately for each region and lead time. And importantly, that threshold is calculated only on training data, and then frozen.

▸ *POINT TO THE MAP*

The map on the right is what a forecaster would actually receive — a confidence map showing where the forecast is more likely to fail. And this is one example date; it demonstrates the product, not overall performance.

And this is not only a picture. Ye ek working console hai — a browser dashboard and a read-only API, showing confidence for every gridpoint at every lead day, and the reasons behind each one. Prototype hai, locally chalta hai.

▸ *POINT TO THE PANEL ON THE LEFT — the three reasons under "Flagged because". That panel is Expected Outcome 4 of the problem statement, explainable output; the dashboard and API are Outcome 5. Say "prototype", and say it runs locally — it is not a public service.*

▸ *PAUSE*

Now the obvious question is — is VISHWAS simply learning the rainfall forecast?

Humne rainfall forecast ko completely remove karke test kiya.

▸ *BIG PAUSE*

Performance did not fall. It actually increased, from 0.728 to 0.731.

▸ *STOP. Do not introduce Paras. Look at Mahika — she bridges into Slide 4.*

SLIDE 6 — CLOSING CARD

▸ *Dhruv ends on "…or to our own logs." Take the OUR OWN MEASUREMENT card immediately.*

And briefly, this is the scale behind those results.

Our baseline covered four seasons and three years, with 77,425 wet samples across 1,252 gauge-observed gridpoints.

▸ *PAUSE*

For VISHWAS, we trained on 2018 and tested on 2019, with 651,040 rows in the test set, from around 1.3 million labelled rows overall.

▸ *PAUSE*

And uncertainty bhi hum individual rows resample karke measure nahi karte. We resample entire forecast dates, rather than treating every gridpoint as an independent observation.

So the result is not just a model score — it comes from an evaluation designed to make the comparison fair.

▸ *STOP completely. Look at Mahika. Do not say thank you and do not invite questions — she closes.*

6 EVERY NUMBER YOU SAY, AND WHERE IT COMES FROM

If a judge challenges a figure, give the source — not a defence.

CLAIM	SOURCE / WHAT IT REALLY MEANS
43.9% of the variance	Four-season baseline measurement, CLAUDE.md. Pooled spread–error relationship.
under 1%	0.17%–0.90%, the range spanning five different definitions of a 'wet' gridpoint — so the finding does not rest on one arbitrary cut.

CLAIM	SOURCE / WHAT IT REALLY MEANS
0.565 vs 0.728	Stratified ROC-AUC. ablations/a_full.log.
+0.13 to +0.16	95% interval entirely above zero, cluster bootstrap over the 52 forecast dates. If pressed for the interval say +0.108 to +0.193 — that is what the figure on slide 4 prints. Never give the margin as a single number.
a test year the model had never seen	Trained on 2018 (39 dates fitted, 13 held as validation), tested on all 52 Wednesdays of 2019.
1,210 GRIB messages, 468 MB per date	docs/TECHNICAL-REFERENCE.md §3. Byte-range reads via the .idx sidecar — we never pull a whole file.
50.3 GB, 104 forecast dates, 1.3 million gridpoint-days	Bulk run completed 29 Aug, 2018–2019. 1,252 points × 10 lead days = 12,520 rows per date.
23 features	models/vishwas_model.joblib stores 23 feature names; ablations/permutation_importance.csv has 23 rows. The feature table has 24 non-key columns, but `season` is built and never fed to the model.
95th percentile, per region and lead time	06_label.py. Train-window only, then frozen and applied to test.
31 July 2019, Day 5, valid 5 August	Computed from data/model_predictions.parquet. 454 of 1,252 busts on that date, AUC 0.886. An earlier build had this date wrong by five days — it is right now, and 31 July 2019 was a Wednesday, which every forecast in this study is.
0.728 → 0.731	Ablation c_no_precip. Removing the rainfall forecast raises the score.
77,425 wet samples, 1,252 gridpoints	README.md results table.
651,040 test rows of 1,302,080	The test set is half the labelled table. Do not say '0.728 from 1,302,080 rows' — that conflates the two, and it was a defect we fixed.
cluster bootstrap over dates	The independent unit is the forecast, not the gridpoint.

7 QUESTIONS THE JUDGES WILL ASK YOU

Yours to answer. Read them out loud once — an answer you have said aloud comes back under pressure; one you have only read does not.

Q1 **How do you know you are not just predicting where it rains?**

Three answers, in order of strength.

One — that is exactly what the stratified evaluation controls for. The headline number is measured within rainfall bands, never pooled.

Two — we tested it directly. We removed the rainfall forecast from the model entirely and the score went up, 0.728 to 0.731.

Three — busts genuinely do happen where it rains. The bust rate is about 41% in the wettest of the eight rainfall bins against 0.15% in the driest. So the honest test is whether we beat 'it is raining' alone, and within each band we score 0.728 against 0.565.

⚠ This is the most likely question in the room. Say 'eight bins', not 'deciles'.

Q2 Isn't this just gradient boosting doing gradient boosting things?

We tested that. A model given only ensemble spread scores 0.562, against 0.565 for spread used as a plain ranking. The algorithm adds nothing.

All of the edge comes from having more information — the ensemble's disagreement across moisture, instability, the mid-level pressure pattern and the winds, not just its rainfall disagreement.

Q3 What is your accuracy?

We do not report accuracy, and that is deliberate. The base rate is about 6%, so a model that always says ‘no bust’ scores 94% and is useless.

We report ROC-AUC within rainfall regime, PR-AUC lift over the base rate, and operational POD, FAR, CSI and ETS at a matched alert budget.

⚠ If a judge quotes that 94% back at you, be clear it is the do-nothing model's number, not ours. It is the only accuracy figure anyone on this team should say aloud, and only ever as the thing we are arguing against.

Q4 What does 0.728 actually mean?

ROC-AUC. Take a forecast that busted and one that did not, at random, from within the same rainfall band. 0.728 is the probability we score the bust higher. 0.5 is a coin flip. The current method gets 0.565.

Q5 One test year is thin.

Agreed, and we say so on the slide. The interval resamples 52 dates inside a single year, so it is blind to year-to-year variation — treat it as a floor, not a ceiling.

Extending to five years needs no new code and no re-downloading. That is mid-September work.

Q6 What does it score without the stratification?

0.909 against 0.885. And the point of that pair is the gap: only +0.023, because pooling lets ensemble spread track rainfall magnitude for free and inherit skill it never earned. That is exactly why we do not headline it.

⚠ ONLY if a judge asks directly. Never volunteer it. It appears on no slide and in no figure by deliberate decision.

Q7 Is the output calibrated?

Yes — isotonic regression, fitted on the validation slice. Brier 0.04217 against 0.04749 for spread.

The component that matters is resolution, which measures informativeness, and ours is 1.4 times the incumbent's.

The raw model was already well calibrated in aggregate — I would not claim a rescue. What isotonic fixes is the tail: the raw model said 90% or higher on 72 cases of which only 5.6% actually busted, so isotonic caps the output at 0.75. The calibrated system declines to claim a confidence it has never earned on held-out data.

Q8 What are the 23 features?

Mid-level pressure pattern at 500 hPa and its gradient and flow deformation; winds at 500 and 850 hPa; true vertical shear between those two levels; precipitable water; CAPE; the ensemble's own rainfall forecast; anomalies of moisture, instability and rainfall against a training-only climatology; month; and lead day.

And critically, the spread — the member-to-member disagreement — on geopotential, winds, moisture, instability and rainfall. The model does not only see what the atmosphere is doing. It sees where the ensemble cannot agree with itself, and it learns which disagreements matter.

⚠ If pressed on shear: we could originally only compute horizontal deformation at one level, which is a real predictor but is NOT vertical shear. Rather than mislabel it we fetched a second pressure level at a 60% increase in data transfer and kept both features under honest names.

Q9 Why HistGradientBoosting and not a neural network?

Tabular data, 1.3 million rows, 23 features, CPU only, and the problem statement requires explainable output — so we need per-prediction attribution, which TreeSHAP gives us directly.

And as I said, a model given only spread scores 0.562. This is not a question about model class.

Q10 How did you pick the 95th percentile?

It was fixed before we saw any results. It gives roughly a 5% base rate in training by construction — and the test base rate is deliberately not exactly 5%, which is the proof the split is honest. If the threshold had seen test data, every test row would have helped set the standard it is judged against.

Q11 What if the model had not beaten the baseline?

We would have reported it. The decision rule was fixed before results were seen, and it is written into the project's own rules: if we cannot beat ensemble spread, say so.

And the measurement of the incumbent stands on its own. That the current method explains under 1% of the variance in forecast error, once you compare within rainfall regime, is a finding regardless of what our model does.

Q12 SHAP — what actually drives it?

SHAP and permutation importance pick the same top three — precipitation spread, forecast rainfall, and precipitable water — but they rank them differently, so I will not claim they agree on an ordering.

The more interesting result is in individual cases. There are forecasts where the ensemble predicted 0.0 millimetres, spread was exactly zero, we flagged anyway, and 13 to 22 millimetres fell. In those cases zero spread is the largest negative contributor — the model correctly learned that zero spread argues against a bust, and flagged anyway because the moisture signal overwhelmed it. That is a system overruling the incumbent using information the incumbent does not have.

Q13 Why only Wednesdays?

A property of the archive, not a choice we would have made. GEFS version 12 reforecast carries 11 ensemble members on Wednesdays and only 5 on other days.

We take the 11-member days because member count is exactly what the baseline depends on. It also gives clean weekly sampling with no day-of-week confound.

Q14 **50 gigabytes. What did you run that on, and how long did it take?**

A laptop, about three hours at roughly 4.8 megabytes a second for 104 forecast dates.

The fetch is resumable — it survives being killed mid-download — and it streams each date rather than holding the whole archive. At 468 megabytes a date, a fetch that had to restart from zero would not have finished in the time we had.

Q15 **How do you colocate a 0.5 degree forecast grid onto 0.25 degree gauge data?**

Exact colocation onto our 0.5 degree analysis grid, and it is verified independently — a separate script with four checks, all passing.

Ocean and no-gauge cells arrive as minus 999 and are masked out, which is what leaves us 1,252 of 5,041 gridpoints.

Q16 **Why 0.5 degrees? Why not finer?**

The analysis grid is 5 to 40 north, 65 to 100 east, at 0.5 degrees — 71 by 71, so 5,041 points before the land mask.

That matches the scale at which a medium-range ensemble actually carries information. Going finer would be resolving detail the forecast does not have.

Q17 **Could this run on live forecasts tomorrow?**

The features come from ensemble fields any operational centre already produces, and it trains and runs on CPU. It is a layer over an existing run, not a new run.

What we would need is access to the live feed — our training sits entirely inside the 2000 to 2019 reforecast archive today, and we take no real-time observations.

Q18 **Why not use IMD's automatic weather station data?**

Public portal access to the AWS and ARG station data closed around May 2025. The gridded gauge product is the one that is available and citable.

Q19 **Is it reproducible?**

Every stage is a numbered script, the model artifact is persisted and reproduces its predictions exactly, and all five ablations are logged. Both datasets are public and free.

Q20 **Only 11 ensemble members? Operational systems run 31.**

Tested. Going from 5 to 11 members improves the baseline slightly and does not change the conclusion — which makes our own claim marginally weaker, and we wrote it down anyway. 11 to 31 cannot be measured from this archive, and we say that too.

8 IF IT IS NOT YOURS, HAND IT OVER

Naming the right teammate reads as a team that knows its own work. Guessing reads as one that does not.

IF THEY ASK ABOUT…	SAY THE NAME
What an ensemble is, why gauges rather than reanalysis	Ankit
A real-world example of a bust	Arnav
Metric definitions, the alert budget, the defects we fixed	Paras

IF THEY ASK ABOUT…	SAY THE NAME
Viability, the risks, why GEFS and not NEPS	Arnav
What users said	Mahika
Sources, impact, deployment path	Dhruv

SITUATION	WHAT TO SAY
You don't know the answer	“I don't have that number to hand and I'd rather not guess — can I come back to you on it?” That is a strength for this team, not a weakness: when we could not trace one of our own figures, we removed it rather than present it.
It's a teammate's area	“That's <name>'s part — <name>, do you want to take that?” Say the name. Don't answer for them.
A judge challenges a number	Give the source, not a defence. Every figure on the deck traces to a government source, a peer-reviewed paper, or our own logs.
A judge says 'you're just predicting heavy rain'	Anyone can answer this one: we removed the rainfall forecast from the model entirely and the score went UP, 0.728 to 0.731.

9 NEVER SAY THIS

Every one of these was a real error caught during the build, or a rule set deliberately.

- × Never say '43.9% of forecast error'. It is 43.9% of the VARIANCE in forecast error. This is the single easiest thing to get wrong under pressure.
- × Never quote the margin as '+0.163'. Say '+0.13 to +0.16'.
- × Never say 'deciles' or 'the wettest tenth' about the model result. It is EIGHT bins. (Deciles are correct only for the separate Phase 1 correlation analysis.)
- × Never quote any accuracy figure. Base rate is about 6%, so accuracy is meaningless — and quoting it signals a team that does not understand imbalance.
- × Never say 'AI-adjusted forecast' or imply we correct the forecast. We DETECT busts. We do not correct anything.
- × Never volunteer the pooled ROC-AUC pair (0.909 vs 0.885). It is a Q&A-only answer if a judge asks directly what the crude comparison gives. Putting it forward would commit, in our own work, exactly the error our pitch accuses the incumbent of.
- × Never quote +0.663 as 'the baseline we beat', and never quote a margin range containing +0.035 (that one came from a filter conditioning on the outcome).
- × Never call the regions 'IMD official regions'. They are approximate lat/lon boxes.
- × Never present the confidence map's date as evidence of general performance. It is ONE date, illustrating the form of the product.
- × Never convert the three field interviews into percentages or 'N% of users said'. They are three exploratory conversations.
- × Never quote 98%→46%, ₹13,331 cr, '50% faster', '~30% improvement', or a 0.34–0.37 false-alarm ratio. All were in earlier drafts; none could be traced; all were removed. Do not say any of these numbers out loud even while explaining that you removed them.
- × Never say SHAP and permutation importance 'agree on the drivers'. They pick the same top three in a different order.

10 NUMBER CHEAT SHEET

Same in all six packs. If two of you quote a figure differently in the same room, that is the damage.

WHAT	NUMBER	WATCH OUT
Spread explains, pooled	43.9% of the VARIANCE in forecast error	not '43.9% of forecast error'
Spread explains, fair comparison	0.17%–0.90% → say 'under 1%'	never quote +0.041 alone
VISHWAS score (stratified ROC-AUC)	0.728	—
Ensemble spread score	0.565	—
Margin	+0.13 to +0.16 · 95% CI +0.108 to +0.193	never '+0.163' alone
Matched 10% alert budget	VISHWAS catches 64% of busts vs 55%	—
POD / FAR / CSI / ETS at 10%	0.641 / 0.586 / 0.336 / 0.300 vs 0.546 / 0.647 / 0.273 / 0.235	—
Features	23, all from GEFS, none from the truth	not 24
Ensemble members	11 (control + 10 perturbed)	—
Domain	1,252 of 5,041 gridpoints (24.8%), land only	—
Data fetched	50.3 GB · 104 forecast dates	—
Labelled rows	1,302,080 total · 651,040 in the test set	don't conflate the two
Train / test	trained on 2018 · tested on all 52 Wednesdays of 2019	—
Base rate	about 6%	so never quote accuracy
Bust definition	|forecast − observed| above the 95th percentile, per region and lead time, threshold set on training data only	—
Rainfall bins	EIGHT bins · 40.92% bust rate in the wettest, 0.15% in the driest	never 'deciles' or 'wettest tenth'
Ablations	full 0.728 · no month 0.722 · no rainfall forecast 0.731 · no 850 shear 0.726 · spread-only 0.562	—
Underlying GEFS, heavy rain (≥64.5 mm)	detects 23% of events at Day 1, 0.2% at Day 10	this is the forecast, NOT our model
IMD warning accuracy (MoES)	85, 73, 67, 63, 58% for Day 1–5	Lok Sabha Q.2953, 17 Dec 2025
The tie	ensemble spread is exactly 0.00 mm on 189,758 test rows — a third of the set	—
Calibration	Brier 0.04217 vs 0.04749 · resolution 1.4× the incumbent's	—

ONE LAST THING

The strongest thing about this project is not the score. It is that we measured the method used today before proposing a replacement, and that we removed our own figures when we could not source them. If you land one impression, land that one.